

M2 COMMUNITY NODE

Troubleshooting Guide

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Symptom-based field reference — WiFi · ATAK · Matrix · Mesh · Monero · Tor

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M2 Community Node — Troubleshooting

Organized by symptom. Every entry here was either experienced during the reference build or identified as a high-probability failure mode.

WiFi / Network

WiFi clients can't reach the Pis after changing router settings

Symptom: Phones connect to WiFi but can't load the community page or reach any Pi service. Wired devices work fine.

Cause: GL.iNet firmware silently sets `ap_isolate=1` after any WiFi configuration change. This blocks WiFi clients from reaching wired devices (the Pis) even on the same subnet.

Fix:

```
ssh root@192.168.8.1
hostapd_cli -i wlan0 set ap_isolate 0
hostapd_cli -i wlan1 set ap_isolate 0
```

If you have a boot script that applies this fix, verify it ran:

```
logread | grep ap_isolate
```

Note: This fix does not survive a router reboot unless applied via a scheduled script. The reference build runs this fix at 15s and 30s after boot via a cron job on the router.

Devices won't connect after power cycle — wait 60 seconds

Symptom: Just power-cycled the rack. Nothing works. Phones can't connect.

Cause: The router takes 30–45 seconds to fully boot and apply all scripts. The AP isolation fix scripts run at 15s and 30s — if the router hasn't fully initialized, they fail silently.

Fix: Wait 60 seconds after power-on before testing. Then test from a wired device first, then WiFi.

Can't get static DHCP leases before first boot

Symptom: Trying to assign fixed IPs (192.168.8.10 and 192.168.8.20) to the Pis before they've booted.

Cause: You need the Pi's MAC address to create a static lease, but the Pi hasn't booted yet to reveal its MAC.

Fix: Boot both Pis on DHCP first. Find their assigned IPs in the router's connected devices list. Note their MAC addresses. Then create static leases and reboot the Pis.

DHCP range conflicts with static IPs

Symptom: Router assigns 192.168.8.10 or 192.168.8.20 to a random device.

Fix: Set the DHCP range to start above your static IPs. Example: static leases at .10 and .20, DHCP range starts at .100.

Docker / Services (Pi #1)

`docker compose up` fails with missing variable error

Symptom: Error like variable `CLOUDFLARE_TUNNEL_TOKEN` is not set

Cause: `.env` file missing or incomplete. Docker Compose requires all variables referenced in the compose file to be defined.

Fix:

```
cd /opt/community-node
ls -la .env      # verify it exists
cat .env        # verify all variables are populated
```

Create from the template if missing:

```
cp .env.template .env
nano .env        # fill in all values
```

Conduit (Matrix) won't start — SSL certificate error

Symptom: `docker logs conduit` shows certificate errors. Matrix client can't connect.

Cause: SSL certificates either don't exist yet or are expired.

Fix:

```
certbot renew --dry-run    # test renewal
certbot renew             # renew if dry-run succeeds
docker compose restart nginx conduit
```

Verify cert is valid:

```
openssl s_client -connect matrix.yourdomain.com:443 </dev/null 2>&1 | grep "Verify return code"
```

Expected: Verify return code: 0 (ok)

Conduit admin room commands don't work

Symptom: Commands sent to `@conduit:yourdomain.com` are ignored or return errors.

Cause 1: The admin room is encrypted. Conduit's admin bot cannot participate in E2EE rooms.

Fix: Create a new unencrypted room and invite `@conduit:yourdomain.com`. Use only that room for admin commands.

Cause 2: You're not using the admin account. Only the first registered user (admin) has control.

Element Web loads but can't log in

Symptom: Element Web loads the login screen but authentication fails.

Cause 1: Homeserver URL is wrong. Element is configured to point to `matrix.yourdomain.com` but you're accessing it from a LAN IP.

Fix: Verify `element-config.json` has the correct homeserver URL for your deployment. On LAN, the homeserver is `http://192.168.8.10:6167` or the LAN-accessible Matrix URL.

Cause 2: Conduit is not running.

```
docker compose ps    # check conduit status
docker logs conduit  # check for errors
```

Cloudflare tunnel is not routing traffic

Symptom: External domain (element.yourdomain.com) returns 502 or times out.

Fix:

```
docker logs cloudflared    # look for connection errors
```

If tunnel shows disconnected:

```
docker compose restart cloudflared
```

Verify the tunnel token in .env matches the token in Cloudflare Zero Trust > Networks > Tunnels.

AdGuard Home blocks internal services

Symptom: After enabling AdGuard, node services stop resolving by hostname.

Fix: Add custom DNS rewrites in AdGuard Home (Settings > DNS Rewrites) for your local services:

```
matrix.yourdomain.com → 192.168.8.10
element.yourdomain.com → 192.168.8.10
```

This ensures local resolution goes to the Pi instead of the external IP.

ATAK

ATAK can't connect to OpenTAK Server — connection refused

Symptom: ATAK shows "Connection failed" or times out when trying to reach the server.

Check 1: Is OTS running?

```
ssh pi@192.168.8.20
sudo systemctl status opentakserver
```

Check 2: Is ATAK using the right IP and port?

- IP: 192.168.8.20
- SSL port: 8089
- TCP port (insecure): 8088

Check 3: Is the device on the CommunityNode WiFi? ATAK over LAN only works when on the same network as the node.

Check 4: For remote access via VPN — is the device connected to Headscale? Check in Headscale admin or run `tailscale status` on the device.

ATAK manual enrollment fails — truststore password rejected

Symptom: ATAK asks for truststore password during manual setup and rejects the entered value.

Default password: atakatak

If you changed this during OTS installation, retrieve it from your secrets file. There is no way to recover the truststore password other than from your records — if lost, you must regenerate the truststore in OTS and re-enroll all clients.

Where it's set: OTS admin console > Certificate Enrollment > Truststore settings.

ATAK positions stop updating / COT not flowing

Symptom: Team members are connected but positions don't update on the map.

Diagnosis:

```
ssh pi@192.168.8.20
sudo systemctl status opentakserver
sudo journalctl -u opentakserver -n 50
```

Look for: database connection errors, memory pressure, or COT parser failures.

Common cause: OTS PostgreSQL database ran out of space (Monero sync fills the NVMe, OTS shares it).

```
df -h    # check disk usage
```

If disk is full, the Monero node is the most likely culprit — prune the blockchain or add storage.

Reporting strategy — positions update inconsistently

Symptom: Some team members show real-time positions, others are stale.

Cause: ATAK reporting mode defaults to "Dynamic" which only updates when the device moves. For event use, all EUDs should be set to "Consistent" (updates on a fixed interval regardless of movement).

Fix (per device): ATAK > Settings > Show All > Reporting Strategy > Consistent

This cannot be set server-side; each operator must change it on their device.

Monero

Monerod appears hung — no sync progress

Symptom: `monerod` status shows height that isn't increasing, or `systemd` says the service is running but blockchain is at 0%.

Cause: Monero blockchain sync takes 24–72 hours on a Pi. This is normal. Do not restart `monerod` during sync unless you have confirmed it is actually stuck (no disk activity for 30+ minutes).

Check progress:

```
ssh pi@192.168.8.20
sudo journalctl -u monerod -n 20
```

Look for lines like: Synchronized 2500000/3264185, net hash 3.24 GH/s

If actually stuck:

```
sudo systemctl stop monerod
rm /opt/monero/data/p2pstate.bin    # clears stale peer list
sudo systemctl start monerod
```

Monero RPC returns authentication error

Symptom: Wallet software or status scripts get 401 from `192.168.8.20:18089`.

Cause: RPC digest credentials are wrong. The password is set in `monerod.conf` and must match what your wallet/script uses.

Fix: Check `monerod.conf` for `rpc-login=username:password`. Update your wallet config to match.

Reticulum / Meshtastic

RNode not initializing

Symptom: Reticulum shows no LoRa interface. `insd` logs show interface error.

Check: Is the LEFT Heltec V3 flashed with RNode firmware (not Meshtastic)?

```
nodeconf /dev/ttyUSB0 --info
```

If it shows Meshtastic firmware, you have the devices swapped. LEFT = RNode, RIGHT = Meshtastic.

Meshtastic not appearing in ATAK

Symptom: Meshtastic devices are visible in the Meshtastic app but don't show up in ATAK.

Check: Is the Meshtastic ATAK plugin installed and configured to point to the ATAK server at 192.168.8.20:8089?

The plugin creates a local bridge between Meshtastic and ATAK. Without it, the two systems are independent.

Tor / I2P

.onion addresses not resolving

Symptom: Tor Browser can't reach .onion addresses for Element or Matrix.

Check: Is the Tor container running?

```
docker compose ps tor
docker logs tor
```

Check: Are the hidden service directories populated?

```
ls /opt/community-node/data/tor/
```

Each service should have a hostname file with the .onion address.

Important: If you delete the Tor data directory or lose private_key files, your .onion addresses change permanently. Back up the entire data/tor/ directory.

General Diagnostics

Quick health check (run from field laptop or any SSH session)

```
# Pi #1 (comms)
ssh pi@192.168.8.10 'docker compose -f /opt/community-node/docker-compose.yml ps'

# Pi #2 (tactical)
ssh pi@192.168.8.20 'sudo systemctl status opentakserver monerod'

# Disk usage (both Pis)
ssh pi@192.168.8.10 'df -h /'
ssh pi@192.168.8.20 'df -h /'
```


Service ports quick reference

Service	Host	Port	Protocol
Element Web (LAN)	192.168.8.10	8080	HTTP
Community page	192.168.8.10	8081	HTTP
Matrix (Conduit)	192.168.8.10	6167	HTTP (internal)
Nginx / Matrix TLS	192.168.8.10	443	HTTPS
AdGuard Home	192.168.8.10	3000	HTTP
OTS web	192.168.8.20	8080	HTTP
OTS TCP (ATAK)	192.168.8.20	8088	TCP
OTS SSL (ATAK)	192.168.8.20	8089	TLS
OTS cert enroll	192.168.8.20	8446	HTTPS
ATAK enrollment page	192.168.8.20	8447	HTTPS
Mumble voice	192.168.8.20	64738	TCP/UDP
Monero RPC	192.168.8.20	18089	HTTP (LAN only)
Headscale	192.168.8.10	8080	HTTP (via Docker)
GL.iNet admin	192.168.8.1	80/443	HTTP/HTTPS